SimpleTracking Instructions

SimpleTracking Version 2.0

Last Updated: 10/19/2026 by Clara Shin (clara.shin@wsu.edu)

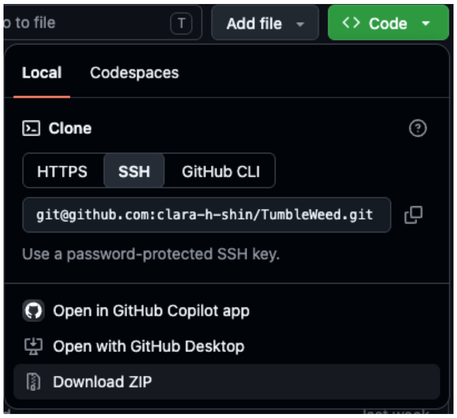
This is the **first step** of chemotaxis analysis in TumbleWeed. `SimpleTracking.m` converts microscopic image sequences of bacterial movements into X and Y coordinates of the trajectories.

- Input: TIFF Image sequences
- Output
  - `tracked_positions.xlsx`: An excel file with X and Y coordinates per track
  - `SimpleTrackingoutput.mat`: This stores key variables and results that will be used for next steps

If you can't find the right setting from `SimpleTracking.m`, we recommend using TrackingGUI for bacterial tracking, and use the resulting `.mat` file from the software for the next step.

Step 0: Download TumbleWeed and Prepare Image sequences

1. Download TumbleWeed from GitHub
   a. Go to https://github.com/clara-h-shin/TumbleWeed
   b. Click < > Code and Download ZIP



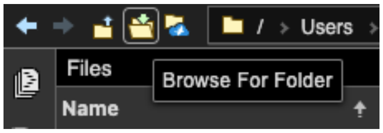
`git@github.com:clara-h-shin/TumbleWeed.git`

Use a password-protected SSH key.

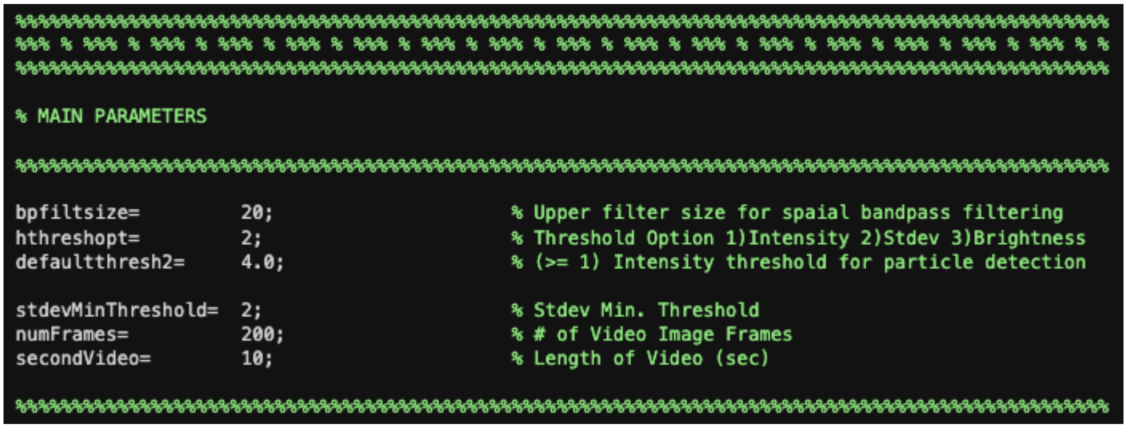
   c. Unzip the file and store where you can easily find!
2. Prepare Image sequences
   a. You will need to have microscopic images/ videos taken
   b. Use ImageJ, Fiji or other imaging software to convert them to TIFF image sequences
   c. See 0_Image_Processing_Instructions.pdf if you need help!

Step 1: SimpleTracking Parameters

1. Open MATLAB and click “Browse For Folder” icon to navigate to the folder that TumbleWeed is downloaded



2. Find and open `SimpleTracking.m`
3. Adjust Main parameters. <u>The most important parameters</u> are `numFrames` and `secondVideo`. Make sure those two parameters are correct. Below is the screenshot of main parameters with default values.



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% MAIN PARAMETERS

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bpfiltsize=         20;                     % Upper filter size for spaial bandpass filtering
hthreshopt=         2;                      % Threshold Option 1)Intensity 2)Stdev 3)Brightness
defaultthresh2=     4.0;                    % (>= 1) Intensity threshold for particle detection

stdevMinThreshold=  2;                      % Stdev Min. Threshold
numFrames=          200;                    % # of Video Image Frames
secondVideo=        10;                     % Length of Video (sec)

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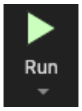
```

4. Adjust other parameters if necessary. All the parameters are pre-set to the setting that is frequently used in the Baylink lab. If you’re unsure about which parameters you should use, see TrackingGUI <u>manual</u> or reach out to us.

Step 2: Select Images

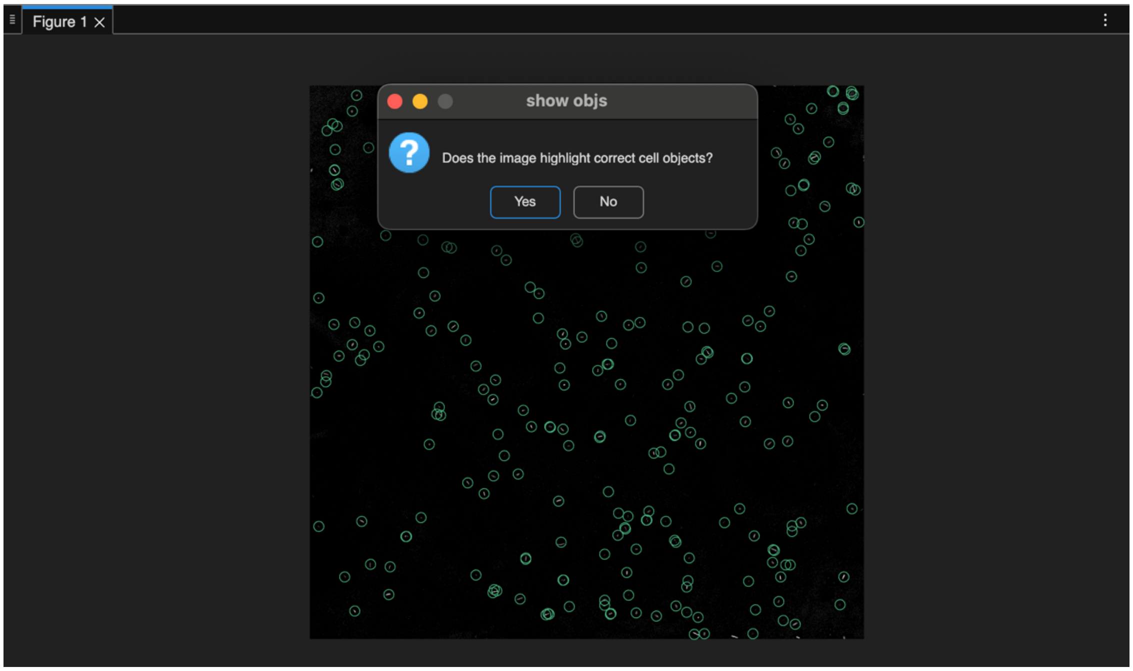
1. Click Run



2. The script will prompt you to select the image files. Browse to the folder with the images
3. Select the first image (format should be `.tif`)
4. Select the last image

Step 3: Confirm Objects

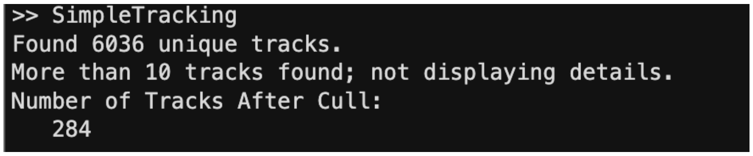
1. The first image of the video will pop up, with green circles on each bacterium object to show that the software will track them.
2. Click “Yes” if the objects look correct. Otherwise, click “No” and the software will stop. Adjust parameters and re-run if incorrect.



Does the image highlight correct cell objects?

Step 4: Track, Link and Cull Objects

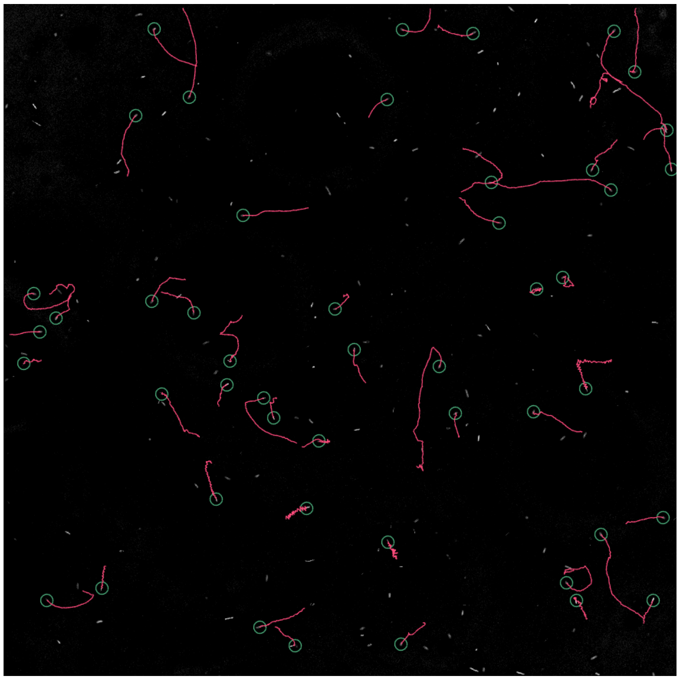
1. After confirming, the software will proceed with tracking, Linking and Culling all objects. You can see the progress with the progress bar.
2. You’ll be able to see the number of tracks after culling on the Command Window. Copy the number if needed. In the screenshot, 6036 is # tracks before the cull and 284 is after.



```
>> SimpleTracking
Found 6036 unique tracks.
More than 10 tracks found; not displaying details.
Number of Tracks After Cull:
   284
```

Step 5: Save Objects

1. Next, another popup will show with the object tracks after the cull with red lines. The green circle is a starting point of a bacterium. Confirm that the tracks look correct.



2. SimpleTracking generates the outputs (`SimpleTrackingoutput.mat`, `Tracked_positions.xlsx`). Save the outputs in a different location or change the name of them if necessary.

References

- Parthasarathy, R. Rapid, accurate particle tracking by calculation of radial symmetry centers. *Nat. Methods* **9**, 724–726 (2012).
  - TrackingGUI GitHub: https://github.com/rplab/TrackingGUI_and_Localization_Public